There is no universal set

The Open Logic Project contributors

openlogic:set-theory/z:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own theorem environment thm:NoUniversalSet (source lines 875-877, the \ollabel on line 875); the proof environment begins two lines below it and is its own proof, closing at source line 883.

Editorial note. Neither the statement nor the proof contains an \olref, an \Olref or a \cref, so no source pointer and no bibliography entry are needed; Separation and Russell's Paradox are both treated in the retained unit and every symbol is declared in the retained preamble.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained theorem prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's Zermelo set-theory unit, counted once: there is no universal set.

Allowed standard prerequisites (editorial).

- Separation. For any set V and any property of sets, the collection of elements of V satisfying the property is a set.. The proof applies Separation to the assumed universal set V and the property x not in x at source lines 880-881.
- Russell's Paradox. The unrestricted collection of all sets x with x not in x leads to a contradiction.. The proof closes the reductio by identifying R with that unrestricted collection and invoking the paradox (source lines 881-882).

Theorem 0.1. *There is no* universal *set, i.e.,* $\{ x \mid x = x \}$ *does not exist.*

Proof. For reductio, suppose V is a universal set. Then by Separation, $R = \{ x \in V \mid x \notin x \} = \{ x \mid x \notin x \}$ exists, contradicting Russell's Paradox. □